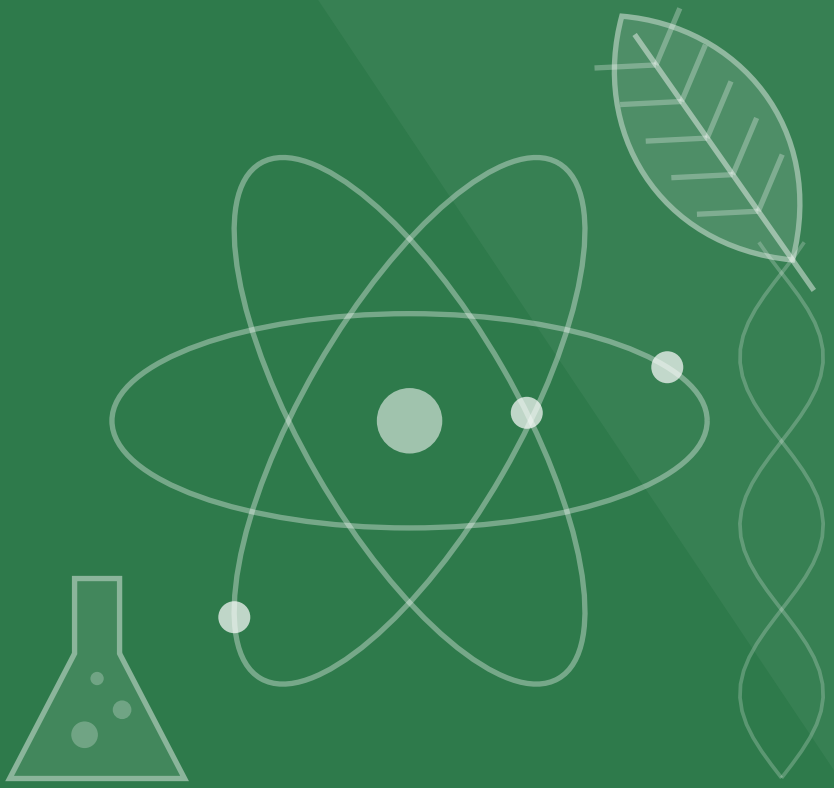
VENVLEARNING

STUDENT TEXTBOOK



Science

GRADE 7

Lessons, examples, and practice for the whole year

2026–2027 School Year · Unit 1: Cycles of Matter and Energy

VENVLEARNING

Grade 7
Science

STUDENT TEXTBOOK

2026–2027 School Year

Unit 1 Cycles of Matter and Energy

This book belongs to _____

A VenvLearning family homeschool curriculum

How to use this book

Each lesson starts with an I can goal and a I Wonder question. Read the lesson with your teacher. Read the Explain It boxes to see how a scientist reasons from evidence, and answer the Check Your Understanding questions. Your practice pages are in your Student Workbook, labeled with the same lesson number.

- Key Idea boxes hold the most important rule or fact. Learn these.
- Watch Out boxes warn you about common mistakes.
- Investigate boxes are hands-on experiments. Do them with your teacher and record what you observe.
- Every Friday there is a short Weekly Check in your workbook. The last Friday of a unit is the Unit Test.

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UNIT 1

Cycles of Matter and Energy

Science practices, Earth's spheres, the water, carbon, oxygen, and nitrogen cycles, photosynthesis, respiration, food webs, and biodiversity.

Every atom in your body has been somewhere else before: in the air, the ocean, a plant, or another animal. In this unit you'll learn how scientists design experiments and explain data, then use those skills to trace how matter cycles and energy flows through Earth's systems and living things.

In this unit you will…

- Identify variables and write CER explanations from data.
- Model the water, carbon, oxygen, and nitrogen cycles.
- Explain photosynthesis and cellular respiration as linked processes.
- Model energy flow in food webs and evaluate ways to protect biodiversity.

SCIENCE IN YOUR WORLD

Farmers, foresters, and environmental engineers use these cycles to grow food, manage forests, and protect water supplies.

UNIT VOCABULARY

independent variable	The variable that is changed.
dependent variable	The variable that is measured.
photosynthesis	Light + CO_2 + water → glucose + oxygen.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
nitrogen fixation	Changing N_2 into usable forms.
biodiversity	The variety of life in an area.

Lessons in this unit

Lesson	Title
1.1	Variables, Data, and Scientific Explanations

Lesson	Title
1.2	Earth's Spheres and the Water Cycle
1.3	The Carbon and Oxygen Cycles
1.4	The Nitrogen Cycle
1.5	Photosynthesis
1.6	Cellular Respiration
1.7	Food Webs and Energy Pyramids
1.8	Biodiversity and Ecosystem Services

UNIT 1 · LESSON 1.1 · SEP 3, 4, 6

Lesson 1.1: Variables, Data, and Scientific Explanations

I can identify independent, dependent, and controlled variables and write a scientific explanation using CER.

WORDS TO KNOW

independent variable	The variable the scientist changes on purpose.
dependent variable	The variable that is measured; it responds to the change.
controlled variables	Conditions kept the same so the test is fair.
hypothesis	A testable prediction that explains how one variable affects another.

I WONDER

Does a fizzing tablet dissolve faster in hot water or cold water? How could you find out fairly?

A controlled experiment tests how one factor affects another. The independent variable is what you change. The dependent variable is what you measure. Everything else stays the same as controlled variables. A good hypothesis is written so it can be tested: “If…, then…, because….”

Part	Fizzing-tablet experiment
question	How does water temperature affect how long a tablet takes to dissolve?
hypothesis	If the water is warmer, then the tablet will dissolve faster, because particles move faster at higher temperatures.
independent variable	water temperature
dependent variable	time to dissolve (seconds)
controlled variables	amount of water, type and size of tablet, cup, no stirring

INVESTIGATE · TEMPERATURE AND DISSOLVING

How does water temperature affect dissolving time?

You need: 3 identical effervescent antacid tablets (or 3 equal spoonfuls of sugar); 3 clear cups; cold, room-temperature, and warm tap water; thermometer; stopwatch.

1. Measure 200 mL of water into each cup and record each temperature.
2. Drop one tablet into each cup at the same moment and start the timer.
3. Record the time when each tablet is completely dissolved.

4 Repeat if you can, and average the trials.

Record: a data table and a line or bar graph with temperature on the x-axis and time on the y-axis.

Safety: Use warm tap water, not boiling water. Do not swallow the tablets or solutions.

EXPLAIN IT · WRITE A SCIENTIFIC EXPLANATION

Data: 8°C → 95 s; 21°C → 52 s; 40°C → 24 s

1 Claim: Tablets dissolve faster in warmer water.

2 Evidence: dissolving time dropped from 95 s at 8°C to 52 s at 21°C to 24 s at 40°C.

3 Reasoning: particles in warmer water move faster, so they collide with the tablet more often and break it apart sooner.

Explanation: A complete explanation connects the evidence to a scientific idea.

KEY IDEA

Change one independent variable, measure the dependent variable, control everything else, and explain results with CER.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Plant height is measured after different amounts of fertilizer. Independent variable?
2. Same experiment: dependent variable?
3. Name two controlled variables.

More practice: Workbook, Lesson 1.1.

LESSON SUMMARY

Independent is changed, dependent is measured, controlled stay the same; explain with CER.

UNIT 1 · LESSON 1.2 · MS-ESS2-4

Lesson 1.2: Earth's Spheres and the Water Cycle

I can describe Earth's four spheres and model how energy from the sun and gravity drive the water cycle.

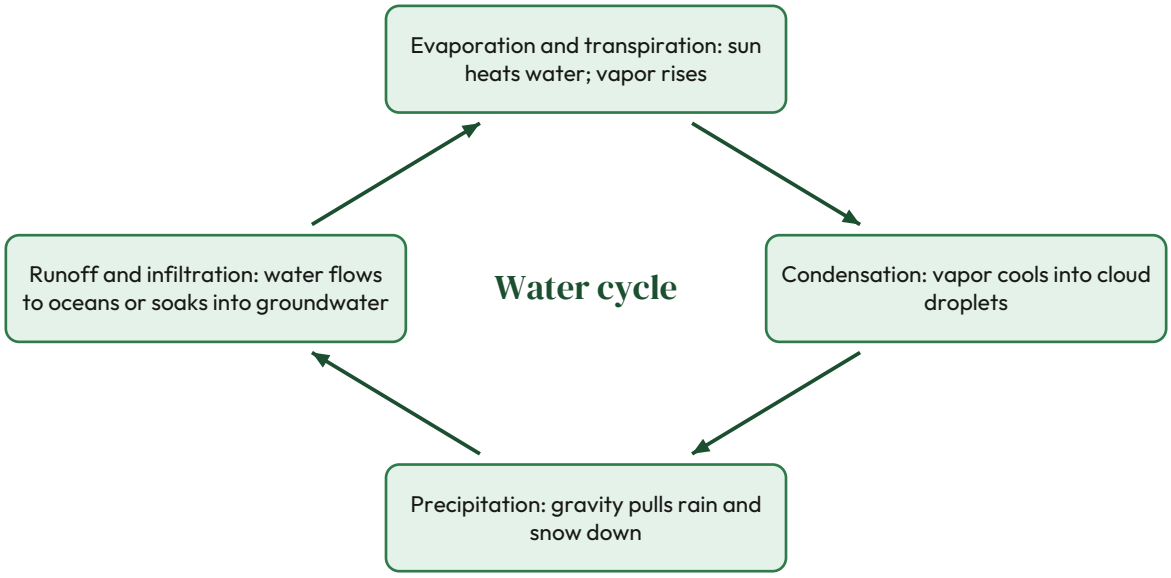
WORDS TO KNOW

geosphere	The solid rock and soil of Earth.
hydrosphere	All of Earth's water.
atmosphere	The layer of gases around Earth.
biosphere	All living things on Earth.
transpiration	The release of water vapor from plant leaves.
condensation	The change from water vapor to liquid water.

I WONDER

The water you drank today may once have been in a dinosaur, a cloud, or an ocean. How could that be?

Scientists study Earth as a system of four interacting parts, or spheres: the geosphere, hydrosphere, atmosphere, and biosphere. Matter, like water, constantly moves among them. Earth has about the same amount of water today as it had long ago; it just keeps cycling.



Two forces drive the cycle. Energy from the sun evaporates water and powers winds. Gravity pulls precipitation down and makes water flow downhill into rivers, lakes, and oceans, and soak into the ground.

INVESTIGATE · WATER CYCLE IN A BAG

Can you model the water cycle?

You need: a zip-top bag; 1/4 cup water with a few drops of blue food coloring; tape; a sunny window; marker.

1. Draw a sun, cloud, and waves on the bag.
2. Pour in the water and seal the bag.
3. Tape it to a sunny window.
4. Observe it after 1 hour, after 1 day, and after 2 days.

Record: observations and a labeled sketch showing evaporation, condensation, and precipitation in the bag.

KEY IDEA

Water cycles among Earth's spheres, driven by energy from the sun and by gravity.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Which sphere includes clouds and air?
2. What process returns water vapor from plants?
3. What force makes rain fall and rivers flow?

More practice: Workbook, Lesson 1.2.

LESSON SUMMARY

Water cycles among Earth's spheres, powered by the sun and gravity.

UNIT 1 · LESSON 1.3 · MS-LS2-3

Lesson 1.3: The Carbon and Oxygen Cycles

I can explain how carbon and oxygen cycle among living things, the atmosphere, the oceans, and rocks.

WORDS TO KNOW

carbon cycle	The movement of carbon among the air, living things, oceans, and rocks.
decomposer	An organism that breaks down dead matter and returns materials to the environment.
fossil fuel	Coal, oil, or natural gas formed from ancient living things.
combustion	Burning; a reaction with oxygen that releases energy and carbon dioxide.

I WONDER

Carbon is in your body, in a diamond, in soda bubbles, and in the air. How can one element be in all of these?

Carbon is the building block of living things. Carbon atoms are never used up; they cycle. Plants take carbon dioxide from the air during photosynthesis and build sugars. Animals eat plants. Living things release carbon dioxide during cellular respiration, and decomposers release it as they break down dead matter. Oceans absorb and release carbon dioxide too. Over millions of years, some carbon becomes buried as fossil fuels; burning them releases that carbon quickly.

The oxygen cycle is linked to the carbon cycle: photosynthesis releases oxygen, and respiration and combustion use oxygen. Today air is about 21 percent oxygen and about 0.04 percent carbon dioxide.

Process	Carbon dioxide	Oxygen
photosynthesis	taken in	released
cellular respiration	released	taken in
decomposition	released	taken in
combustion (burning fuels)	released	taken in
ocean absorption	taken in (dissolves in water)	—

INVESTIGATE · FOLLOW A CARBON ATOM

What paths can one carbon atom take?

You need: station cards: Atmosphere, Plant, Animal, Decomposer, Ocean, Fossil Fuel; one die.

1. Start at the Atmosphere station.
2. Roll the die; each station card tells where a roll sends you and why (e.g. 1–3: a plant takes you in during photosynthesis).
3. Record each move for 10 rolls.
4. Compare paths with a partner.

Record: your atom's path as a flow chart, with the process that moved it at each arrow.

KEY IDEA

Photosynthesis removes carbon dioxide and adds oxygen; respiration, decomposition, and combustion do the opposite.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Which process removes carbon dioxide from the air?
2. Name two processes that release carbon dioxide.
3. About what percent of air is oxygen?

More practice: Workbook, Lesson 1.3.

LESSON SUMMARY

Carbon and oxygen cycle through photosynthesis, respiration, decomposition, combustion, and oceans.

UNIT 1 · LESSON 1.4 · MS-LS2-3

Lesson 1.4: The Nitrogen Cycle

I can explain how nitrogen moves from the air into living things and back.

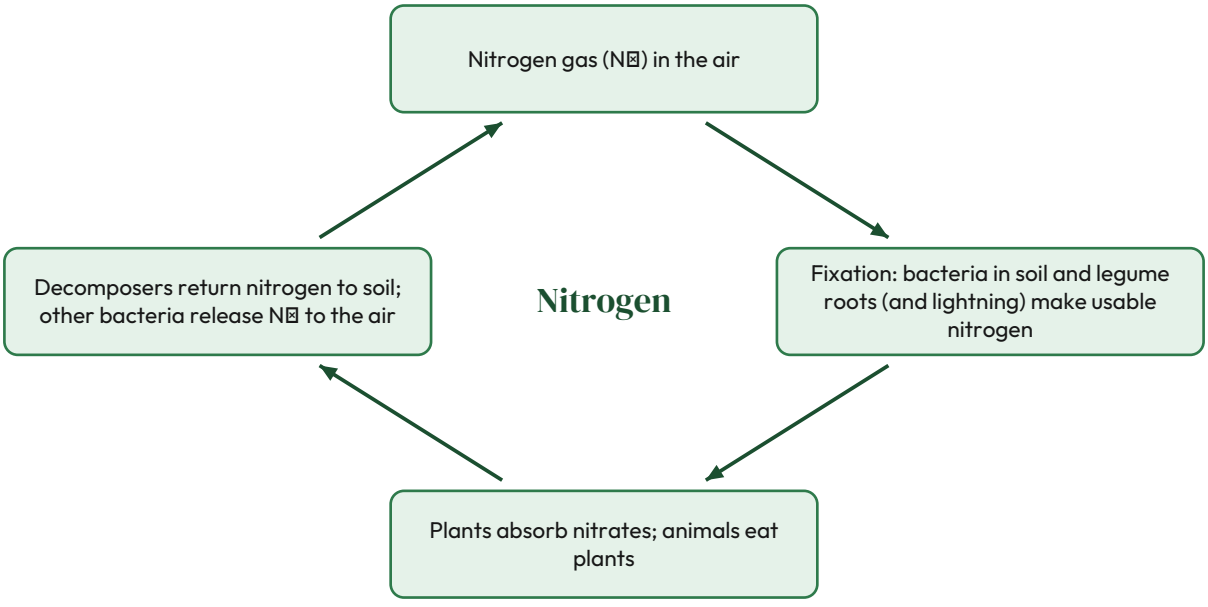
WORDS TO KNOW

nitrogen fixation	Changing nitrogen gas into a form plants can use.
legume	A plant such as a bean, pea, or peanut that hosts nitrogen-fixing bacteria in its roots.

I WONDER

Air is about 78 percent nitrogen, and your body needs nitrogen to build proteins. Why can't you just breathe it in and use it?

Living things need nitrogen to build proteins and DNA, but most organisms can't use nitrogen gas (N_2) directly. It has to be fixed into compounds such as ammonia and nitrates.



Farmers use this cycle. Planting legumes such as soybeans or peanuts adds usable nitrogen to the soil. Peanut farmer and scientist George Washington Carver taught Southern farmers to rotate cotton with peanuts and other legumes to restore worn-out soil.

INVESTIGATE · ROOT NODULE SEARCH

Where do nitrogen-fixing bacteria live?

You need: a bean or pea plant (or photos from your teacher); magnifying glass; water.

1 Gently dig up the plant and rinse the soil from its roots.

2 Look for small round bumps (nodules) on the roots.

3 Sketch the roots and nodules.

Record: a labeled sketch and an explanation of what the bacteria in the nodules do.

KEY IDEA

Bacteria fix nitrogen gas into usable forms; plants absorb it, animals eat plants, and decomposers and bacteria return it.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. What percent of air is nitrogen?
2. What does nitrogen fixation do?
3. Why do farmers plant legumes?

More practice: Workbook, Lesson 1.4.

LESSON SUMMARY

Bacteria fix nitrogen so plants and animals can use it; decomposers return it.

UNIT 1 · LESSON 1.5 · MS-LS1-6

Lesson 1.5: Photosynthesis

I can explain how plants use light energy to make sugar and oxygen from carbon dioxide and water.

WORDS TO KNOW

photosynthesis	The process plants use to turn light energy, carbon dioxide, and water into sugar and oxygen.
chloroplast	The part of a plant cell where photosynthesis happens.
chlorophyll	The green pigment that captures light energy.
glucose	A sugar made in photosynthesis that stores energy.

I WONDER

A huge oak tree grows from a tiny acorn. Where does all that mass come from?

Surprisingly, most of a tree's mass comes from the air! In photosynthesis, chloroplasts in leaf cells use light energy captured by chlorophyll to combine carbon dioxide from the air with water from the soil. They produce glucose, which the plant uses for energy and to build its body, and release oxygen.

Word equation	carbon dioxide + water —(light energy)→ glucose + oxygen
Chemical equation	$6CO_2 + 6H_2O \xrightarrow{\text{(light)}} C_6H_{12}O_6 + 6O_2$

Photosynthesis changes light energy into chemical energy stored in sugar. Almost every food chain on Earth begins with this process.

INVESTIGATE · FLOATING LEAF DISKS

Does light intensity affect the rate of photosynthesis?

You need: fresh spinach leaves; a straw or hole punch; a plastic syringe (no needle); baking soda solution (1/8 tsp in 300 mL water) with 1 drop dish soap; 2 clear cups; a lamp.

1. Punch 20 disks from the leaves.
2. Put 10 disks in the syringe with solution, push out the air, cover the tip, and pull the plunger to remove gas from the leaves until the disks sink. Repeat for the other 10.
3. Put 10 sunk disks in a cup under the lamp and 10 in a cup in a dark place.
4. Count floating disks every minute for 15 minutes.

Record: a line graph of floating disks vs time for both cups, and a CER explaining the difference (oxygen produced in photosynthesis makes disks float).

KEY IDEA

Photosynthesis: carbon dioxide + water + light energy → glucose + oxygen, in chloroplasts.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Where does most of a tree's mass come from?
2. What are the products of photosynthesis?
3. What pigment captures light?

More practice: Workbook, Lesson 1.5.

LESSON SUMMARY

Plants turn carbon dioxide and water into glucose and oxygen using light.

UNIT 1 · LESSON 1.6 · MS-LS1-7

Lesson 1.6: Cellular Respiration

I can explain how cells release energy from food through cellular respiration.

WORDS TO KNOW

cellular respiration	The process cells use to release energy from glucose using oxygen.
mitochondria	Cell parts where cellular respiration releases energy.
fermentation	A process that releases energy from sugar without oxygen.

I WONDER

You ate breakfast hours ago. How is that food powering your brain right now?

All organisms, including plants, need energy. In cellular respiration, mitochondria in cells break down glucose with oxygen, releasing energy the cell can use. Carbon dioxide and water are given off as waste.

Word equation	glucose + oxygen → carbon dioxide + water + energy
Chemical equation	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ + energy

Notice that respiration is roughly the reverse of photosynthesis. The products of one are the reactants of the other, so matter cycles between plants and animals while energy flows from

the sun, through living things, and is eventually lost as heat.

Some organisms, like yeast, can also release energy without oxygen through fermentation, giving off carbon dioxide. That's why bread dough rises.

INVESTIGATE · YEAST BALLOONS

Does yeast need sugar to release carbon dioxide?

You need: 2 empty water bottles; 2 balloons; 2 packets of dry yeast; warm water; 2 tablespoons sugar.

1. Add 1 cup warm water and 1 packet yeast to each bottle.
2. Add sugar to bottle A only. Swirl both.
3. Stretch a balloon over each bottle's mouth.
4. Measure each balloon's circumference every 10 minutes for 30 minutes.

Record: a data table and a graph; explain what filled the balloon and why one grew more.

KEY IDEA

Respiration: glucose + oxygen → carbon dioxide + water + energy, in mitochondria. It is the reverse of photosynthesis.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Where does cellular respiration happen?
2. What gas filled the balloon?
3. Do plants carry out respiration?

More practice: Workbook, Lesson 1.6.

LESSON SUMMARY

Cells release energy from glucose with oxygen; respiration and photosynthesis cycle matter.

UNIT 1 · LESSON 1.7 · MS-LS2-3

Lesson 1.7: Food Webs and Energy Pyramids

I can model how energy flows through a food web and explain why less energy reaches each higher level.

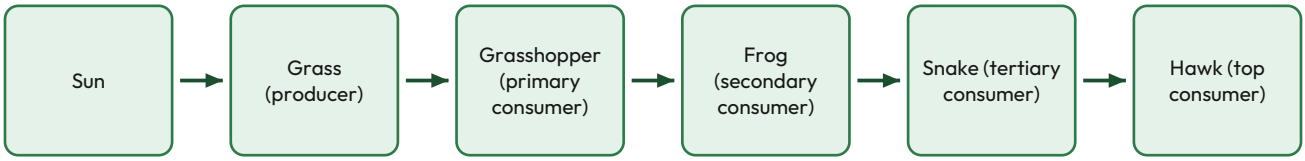
WORDS TO KNOW

producer	An organism that makes its own food, usually by photosynthesis.
consumer	An organism that eats other organisms.
food web	A diagram of all the feeding relationships in an ecosystem.
trophic level	A feeding level in a food chain or energy pyramid.

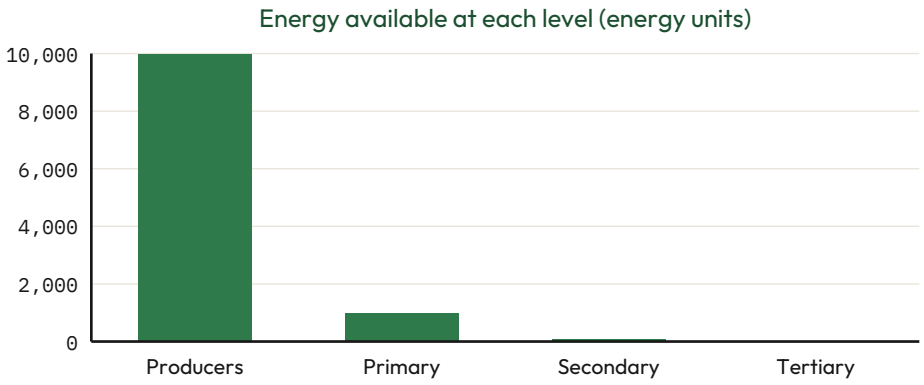
I WONDER

Why are there far more grasshoppers than hawks in a field?

Energy enters most ecosystems as sunlight captured by producers. Primary consumers (herbivores) eat producers, secondary consumers eat herbivores, and so on. Decomposers break down dead organisms at every level. Arrows in a food chain point in the direction energy flows: from the eaten to the eater.



At each level, organisms use most of their energy to live, move, and stay warm, and much is lost as heat. On average only about 10 percent of the energy at one level is stored in the next. That's why an energy pyramid narrows at the top and why there are fewer top predators.



INVESTIGATE · BUILD A FOOD WEB

How are organisms in a forest ecosystem connected?

You need: cards: oak tree, grass, acorn weevil, grasshopper, mouse, rabbit, deer, frog, snake, hawk, fox, mushroom (decomposer); yarn or a pencil.

1 Arrange the cards on a large sheet of paper.

2 Draw arrows from each food to the organism that eats it.

3 Label producers, consumers, and the decomposer.

4 Remove the mouse. Predict two effects on the web.

Record: your food web and your predictions.

KEY IDEA

Energy flows from producers to consumers; only about 10 percent passes to each higher level, while matter is recycled by decomposers.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. Which way do food-chain arrows point?
2. If grass has 20,000 units, about how much reaches grasshoppers?
3. Why are top predators few?

More practice: Workbook, Lesson 1.7.

LESSON SUMMARY

Energy flows up through feeding levels with about 10 percent passed on; matter recycles.

UNIT 1 · LESSON 1.8 · MS-LS2-5

Lesson 1.8: Biodiversity and Ecosystem Services

I can explain why biodiversity matters and evaluate ways to protect ecosystem services.

WORDS TO KNOW

biodiversity	The variety of living things in an area.
ecosystem services	Benefits people get from healthy ecosystems, such as clean water and pollination.
pollinator	An animal, such as a bee, that carries pollen between flowers.

I WONDER

What would happen to your breakfast if all the bees disappeared?

Biodiversity is the variety of species, genes, and ecosystems in a place. Ecosystems with high biodiversity are usually more stable: if one species declines, others can fill its role.

Ecosystem service	Example
pollination	Bees, butterflies, and bats pollinate many fruits, nuts, and vegetables.
clean water	Wetlands filter pollutants as water flows through them.
flood control	Wetlands and forests soak up heavy rain like sponges.
healthy soil	Decomposers and earthworms recycle nutrients.
food and materials	Fish, timber, and medicines come from ecosystems.

EXPLAIN IT · EVALUATE A DESIGN SOLUTION

A town wants to reduce flooding. Option A: build a concrete channel. Option B: restore a wetland upstream.

1. Criteria: reduce flooding, cost, protect biodiversity, clean water.
2. A: moves water fast but removes habitat and doesn't filter water.
3. B: soaks up and slows water, filters pollutants, and provides habitat, but needs land.

Explanation: Option B meets more criteria, especially biodiversity and clean water, though land and cost are constraints.

INVESTIGATE · BIODIVERSITY SURVEY

How many kinds of living things live in a small area?

You need: 4 sticks and string to mark a 1 m × 1 m square; magnifying glass; notebook.

1 Mark a square in a lawn and another in a less-mowed area (garden edge, field, or woods).

2 Count the different kinds of plants and animals in each square for 10 minutes.

3 Compare the two squares.

Record: a table of species counts and an explanation of which area has more biodiversity and why.

KEY IDEA

Biodiversity keeps ecosystems stable and provides services people depend on; good design solutions protect it.

CHECK YOUR UNDERSTANDING

Answer in your science notebook in complete sentences. Then check with your teacher.

1. What is biodiversity?

2. Name two ecosystem services.

3. Why is a diverse ecosystem more stable?

More practice: Workbook, Lesson 1.8.

LESSON SUMMARY

Biodiversity stabilizes ecosystems and provides services worth protecting.

UNIT 1 REVIEW

Unit 1 Study Guide

Use this page to get ready for the unit test. Read each big idea, then try the review pages in your workbook.

Lesson	Remember...
1.1	Independent is changed, dependent is measured, controlled stay the same; explain with CER.
1.2	Water cycles among Earth's spheres, powered by the sun and gravity.
1.3	Carbon and oxygen cycle through photosynthesis, respiration, decomposition, combustion, and oceans.
1.4	Bacteria fix nitrogen so plants and animals can use it; decomposers return it.
1.5	Plants turn carbon dioxide and water into glucose and oxygen using light.
1.6	Cells release energy from glucose with oxygen; respiration and photosynthesis cycle matter.
1.7	Energy flows up through feeding levels with about 10 percent passed on; matter recycles.
1.8	Biodiversity stabilizes ecosystems and provides services worth protecting.

Glossary

atmosphere	The layer of gases around Earth.
biodiversity	The variety of life in an area.
biosphere	All living things on Earth.
carbon cycle	The movement of carbon among the air, living things, oceans, and rocks.
cellular respiration	Glucose + oxygen → CO_2 + water + energy.
chlorophyll	The green pigment that captures light energy.
chloroplast	The part of a plant cell where photosynthesis happens.
combustion	Burning; a reaction with oxygen that releases energy and carbon dioxide.
condensation	The change from water vapor to liquid water.
consumer	An organism that eats other organisms.
controlled variables	Conditions kept the same so the test is fair.
decomposer	An organism that breaks down dead matter and returns materials to the environment.
dependent variable	The variable that is measured.
ecosystem services	Benefits people get from healthy ecosystems, such as clean water and pollination.
fermentation	A process that releases energy from sugar without oxygen.
food web	A diagram of all the feeding relationships in an ecosystem.
fossil fuel	Coal, oil, or natural gas formed from ancient living things.
geosphere	The solid rock and soil of Earth.
glucose	A sugar made in photosynthesis that stores energy.
hydrosphere	All of Earth's water.
hypothesis	A testable prediction that explains how one variable affects another.
independent variable	The variable that is changed.
legume	A plant such as a bean, pea, or peanut that hosts nitrogen-fixing bacteria in its roots.
mitochondria	Cell parts where cellular respiration releases energy.
nitrogen fixation	Changing N_2 into usable forms.
photosynthesis	Light + CO_2 + water → glucose + oxygen.

pollinator	An animal, such as a bee, that carries pollen between flowers.
producer	An organism that makes its own food, usually by photosynthesis.
transpiration	The release of water vapor from plant leaves.
trophic level	A feeding level in a food chain or energy pyramid.

Grade 7 Science

Student Textbook · 2026–2027

A full year of Grade 7 science built around questions, investigations, and evidence. Students observe, measure, model, and explain the natural world, then use what they learn to design solutions to real problems.

This Student Textbook teaches every lesson: the big idea, key vocabulary, models and examples, and a short set of practice problems to check understanding. It pairs with the Student Workbook and the Teacher Edition.

INSIDE THIS YEAR

01 Cycles of Matter and Energy

IN THIS BOOK

- Clear lesson goals: “I can…” statements
- Key vocabulary with student-friendly definitions
- Models, examples, and Key Idea summaries
- A study guide and glossary for every unit

THE COMPLETE SET



Textbook
You are holding this book

Workbook

Teacher Edition

Each lesson has the same number in all three books, and every day of the year is mapped to its lesson in the VenvLearning planner.